

COMPANY BRIEF

Project Helix - Clinical Genomics Quality Engineering Programme
GenomeBridge Clinical Systems Ltd | Version 1.0 | 4 August 2026

1. Who GenomeBridge Is

GenomeBridge Clinical Systems Ltd is a **fictional UK HealthTech company** that builds clinical referral, patient-record and genomic-testing workflow technology used by NHS-facing laboratory and diagnostic partners. It does not process real patient data in any environment used by this programme. All records used in Project Helix are public demonstration data or synthetically generated data.

2. The Ten-Step Clinical Workflow Under Review

1	Locate or register a patient
2	Review patient demographic information
3	Create a clinical referral
4	Record consent
5	Create a genomic test request
6	Associate a specimen with the request
7	Track the processing status
8	Make a diagnostic report available
9	Control access to clinical information
10	Retain a traceable history of material activity

3. Why Now - The Trigger

GenomeBridge support desk (James Wright, Service Support Lead) has logged a sharp rise in complaints over the last quarter tied to steps 1, 6 and 7 of the workflow above: patients being hard to locate, specimens not always linking correctly to service requests, and session interruptions mid-workflow.

The headline figure is unverified. James's team estimates that specimen-to-request linkage problems are now a common cause of delayed genomic test turnaround, but no one has quantified this against actual workflow data. Daniel Brooks (Engineering Manager) has said the team does not trust the current numbers enough to act on them.

This is the candidate's first job: turn an anecdote into evidence.

4. Cost of the Gap

Metric	Value
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Annual cost of unresolved workflow reliability problems	GBP 340,000 (estimated)
Release date	Friday 26 September 2026 - fixed board commitment
P1 open defects permitted at Go/No-Go review	Zero

5. Delivery Timeline

Phase	Focus	Week	Dates
Setup Gate	Workspace, tools, repo, Jira	Pre-work	Before 4 Aug 2026
Phase 1	Quality Discovery, Risk, Test Strategy	Week 1	4 to 8 Aug 2026
Phase 2	Exploratory Testing, Test Design, Defect Triage	Week 2	11 to 15 Aug 2026
Phase 3	FHIR API Quality Engineering	Week 3	18 to 22 Aug 2026
Phase 4	Health Data Quality, SQL, Test-Data Management	Week 4	25 to 29 Aug 2026
Phase 5	Cypress Regression Engineering	Week 5	1 to 5 Sep 2026
Phase 6	Playwright Cross-Browser Engineering	Week 6	8 to 12 Sep 2026
Phase 7	CI/CD, Performance and Quality Signals	Week 7	15 to 19 Sep 2026
Phase 8 + Release Review	Release Readiness and Go/No-Go	Week 8	22 to 26 Sep 2026

6. Project Team

Daniel Brooks - Engineering Manager - candidate's direct manager

Wants faster automated feedback and fewer unreliable tests. Skeptical of automation that passes on retry. Owns the Go/No-Go decision jointly with Maya Patel.

Mark Evans - QA Engineer - outgoing, candidate's predecessor

Built the original Cypress approach with inconsistent standards. Leaving GenomeBridge mid-programme. Handover is deliberately incomplete: stages 1 to 3 documented, everything else has gaps.

Aisha Rahman - Software Engineer - data and API stakeholder

Supports API and browser workflow clarification. Delivers data files during the programme with a pattern buried inside them. Expects the candidate to find it, not be told it.

Laura Okafor - Information Governance Lead - compliance stakeholder

Non-negotiable: synthetic data only, correctly protected. Will raise DCB0129/DCB0160 and UK GDPR in Phase 8 stakeholder challenges. The hardest voice in the room at the release review.

Dr Maya Patel - Product Manager - referenced, not a direct inbox contact

Owns Project Helix requirements. Her requirement documents contain the gaps the candidate must surface in Phase 1.

Sofia Chen - Platform Engineer - referenced, not a direct inbox contact

Reviews GitHub Actions configuration from Phase 7 onward. No secrets in code. No excessive traffic against public systems.

James Wright - Service Support Lead - referenced, not a direct inbox contact

Source of the original complaint themes. His support-ticket framing is what Phase 2 exploratory charters are designed to investigate.

7. Clinical Safety Context - Awareness Level Only

GenomeBridge's workflow is health IT. UK health IT has two companion clinical safety standards that apply to this programme. The candidate is not expected to produce a Clinical Safety Case or act as a Clinical Safety Officer. They are expected to understand, at awareness level, how their quality evidence relates to this framework because Laura Okafor will raise it directly at the Phase 8 release review.

Standard	Who it applies to	Relevance to this programme
DCB0129	Organisations that develop health IT systems (GenomeBridge)	GenomeBridge's obligation as system builder. QA evidence from Phases 3 and 4
DCB0160	Healthcare organisations deploying and using health IT systems	Would apply to any NHS trust or lab partner adopting Project Helix. Out of scope for this programme

Key principle: Software quality evidence supports clinical safety assurance. It does not replace it. A passing test suite is not a clinical safety case.

8. Internal Shorthand Glossary

The operational queue report (used from Phase 3 onward) uses the following internal codes. Candidates who skip this glossary will misread the data.

Code	Meaning	Appears In
REF - INC	Referral Incomplete - mandatory field missing	Phase 3 and 4 operational report
SPEC - UNLINK	Specimen Unlinked - not associated with service request	Phase 3 and 4 - primary metric
CONS - MISS	Consent Missing - no consent for a consent-required referral	Phase 3 and 4
DUP - PAT	Duplicate Patient Candidate - two records flagged	Phase 2 and 4
WF - STALL	Workflow Stalled - no Task status update within expected interval	Phase 3 and 4
TAT	Turnaround Time - referral creation to report availability	Phase 4 and 8 scorecard
QGATE	Quality Gate - internal release checkpoint requiring evidence	Phase 7 and 8
TBC	To Be Confirmed - weekly figure not finalised at reporting time	Phase 3 and 4

9. Repository and Links

All project work is submitted to: <https://github.com/alain-Sortnext/project-helix-quality-engineering>

Artefact	Path in repository
Company brief (this document)	docs/company-brief/
Quality strategy and risk register	docs/quality-strategy/
Requirements and key requirements guide	docs/requirements/
Exploratory charters	manual-testing/exploratory-charters/
FHIR API tests	api-testing/

Database validation	database/
Cypress suite	cypress/
Playwright suite	playwright/
CI/CD workflows	.github/workflows/
Performance tests (k6)	performance/k6/
Release artefacts	release/